

Interfaces and Abstract Classes for
Graduate Certificate
Information Technology – Software Design & Programming

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1. Discussion

The most difficult aspect of this assignment was grasping why you would need to implement an interface. When I first began working on the assignment, it seemed pointless because the `NFLPlayer` and `NBAPlayer` classes essentially had the same fields and method implementations. Why repeat code if everything in each class is going to be the same? After writing everything out and creating unit tests, I realized that interfaces are not meant to reuse code; they are meant to define a contract that each class must follow. Since we had a `Player` interface that declared `getName ()`, `getStats ()`, and `getSport ()` every child class of `Player` had to implement those methods. This allowed us to treat all players as the same type of object, even though they were different types of players.

The abstract class implementation made more sense to me because you are able to share state and actually implement part of the class. Creating fields that both child classes would have, such as name and stats, and then implementing the `getName` method in the abstract class removed the need to repeat yourself in each child class. Each child class only needed to worry about returning to the sport they are in. Once again, showing the main differences between using interfaces vs abstract classes. Interfaces force you to implement certain behavior, while abstract classes can provide you with an implemented structure you can build from.

Writing the unit tests for this assignment further solidified my understanding. By simply declaring the type to be `Player` instead of the concrete class, you are forced to use polymorphism. The abstract class implementation showed how you can reuse code and abstract away things that will be the same across all child classes.

An implementation decision I made was not to make each player's stats something different. I left the stats as a basic integer because I didn't want to worry about creating NFL stats vs NBA stats. I just wanted to focus on learning about inheritance.